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HUJI Applicative Research Report — Agriculture & Food — Week 36, 2026

2 paper(s) · Agriculture & Food · Weekly · Week 36, 2026 · Generated August 31, 2026

■■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

Hebrew University's latest research highlights innovative approaches to enhance food security and quality. One study provides a clear pathway to improve beef shelf-life and appearance through targeted nutritional supplements for livestock, addressing significant waste and market appeal challenges in the meat industry. Concurrently, another breakthrough offers fundamental insights into plant community dynamics under water stress, paving the way for advanced precision agriculture models to boost crop yields in drought-prone regions. These advancements collectively underscore the university's commitment to delivering impactful solutions for a more sustainable and efficient global food system.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

- **Oren Tirosh** — AgriTech, FoodTech
- **Niv DeMalach** — AgriTech

AGRICULTURE & FOOD — RESEARCH HIGHLIGHTS

1. Vitamin E supplementation moderates salt-induced oxidation status in beef. 23/50

"Novel Vitamin E feed additive dramatically extends salted beef shelf-life and maintains premium color."

Main researcher: Oren Tirosh · Oren.tirosh@mail.huji.ac.il.

Institute of Biochemistry, Food Science and Nutrition, The Robert H Smith Faculty of Agriculture, Food, and Environment, The Hebrew University of Jerusalem, Rehovot, Israel. Electronic address: Oren.tirosh@mail.huji.ac.il.

Published in Meat science · 2026-09-01

ABOUT THIS RESEARCH

Researchers found that supplementing calves with high doses of Vitamin E before slaughter protects beef from oxidation and discoloration caused by salt treatments. The study establishes a specific dosage and timeframe to maintain meat quality and color during processing.

COMMERCIAL ANGLE

Development of specialized livestock feed additives or precision nutrition protocols to extend the shelf-life and visual appeal of premium salted beef products.

COMMERCIALISATION METRICS

Novelty 3/10

The research applies a well-known antioxidant (Vitamin E) to a known problem (lipid peroxidation) using standard dosage optimization; it lacks a novel mechanism, a proprietary delivery system, or a groundbreaking discovery.

Commercial Potential 6/10

The study identifies a clear, practical application for enhancing meat shelf-life and quality via a specific dosing regimen, but it lacks high-margin protectability as it utilizes a generic, well-known supplement (Vitamin E).

Market Size 6/10

The target market is the global beef industry, which is vast; however, this is an incremental improvement using a known additive (Vitamin E) rather than a disruptive platform technology.

Tech Readiness 6/10

The research has moved beyond basic proof-of-concept to in vivo optimization of dosing and duration in calves, demonstrating clear industrial applicability. However, it remains at a pilot/experimental stage and requires scale-up validation in commercial feedlots before being market-ready.

IP Strength 2/10

The research applies well-known nutritional concepts (Vitamin E supplementation) to a common problem, lacking a novel molecule, proprietary delivery system, or unique mechanism that would meet the non-obviousness threshold for strong patentability.

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2. Mechanisms behind facilitation-competition transition along rainfall gradients. 10/50

"New insights into plant water competition enable precision agriculture for drought-resistant yield optimization."

Main researcher: Niv DeMalach

Institute of Plant Sciences and Genetics in Agriculture, Robert H. Smith Faculty of Agriculture, Food and Environment, The Hebrew University of Jerusalem, Rehovot 76100001, Israel.

Published in Proceedings of the National Academy of Sciences of the United States of America · 2026 Sep 1

ABOUT THIS RESEARCH

The research examines how plants switch from helping each other to competing for resources as water availability changes. It identifies the specific environmental thresholds that trigger these shifts in plant community dynamics.

COMMERCIAL ANGLE

Development of precision agriculture models and drought-resistant crop spacing strategies to maximize yield in water-stressed regions.

COMMERCIALISATION METRICS

Novelty 3/10

The 'facilitation-competition transition' along rainfall gradients is a well-established fundamental concept in ecology (Stress Gradient Hypothesis). Without an abstract to indicate a novel mechanism or a platform-shifting discovery, the title suggests an incremental study of existing theoretical frameworks.

Commercial Potential 2/10

The title indicates a fundamental ecological study on plant interactions and climate, which lacks a direct path to a protectable product, clinical application, or scalable industrial service.

Market Size 2/10

The title indicates basic ecological research on plant interactions and rainfall, which lacks a clear industrial application or direct addressable commercial market.

Tech Readiness 2/10

The title indicates a fundamental ecological study on theoretical biological mechanisms and environmental gradients, which represents basic academic research with no immediate path to commercial application.

IP Strength 1/10

The title indicates a basic ecological study on plant interactions and environmental gradients, which describes natural phenomena and is therefore neither patentable nor industrially defensible.

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